

Notes: Vijh (RFS, 1994)

S&P 500 Trading Strategies and Stock Betas

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1 Big picture

- **Question.** Do S&P 500 trading strategies - index futures, index options, portfolio insurance, program trading, and basket trades - distort measured stock betas?
- **Core mechanism.** When many investors trade the same basket of S&P 500 stocks at the same time, order imbalances can create temporary price pressure. If this common pressure is embedded in measured market returns, S&P 500 stock covariances with the market are overstated and non-S&P 500 covariances are understated.
- **Competing explanation.** The nonsynchronous-trading hypothesis says S&P 500 trading merely makes S&P 500 stock prices less stale. Higher trading frequency raises measured betas for newly added stocks, but in a way that depends on before-versus-after trading volume.
- **Empirical design.** The paper studies 329 stocks added to the S&P 500 between 1975 and 1989 and 79 matched pairs of seasoned S&P 500 and non-S&P 500 stocks.
- **Main addition result.** Daily betas of added stocks fall by 0.072 during 1975-1979, rise by 0.082 during 1980-1984, and rise by 0.211 during 1985-1989. Weekly betas rise by 0.130 during 1985-1989.
- **Matched-pair result.** In 1985-1989, seasoned S&P 500 stocks have higher daily, weekly, and monthly betas than matched non-S&P 500 stocks by 0.380, 0.279, and 0.125.
- **Key separation test.** Even among stocks for which the nonsynchronous-trading model predicts a beta decrease, betas rise by 0.113. This is the cleanest evidence in favor of price pressure.
- **Magnitude.** Interpreting the 0.113 beta increase as price pressure implies that price pressure accounts for about 8.5 percent of daily variance in the CRSP value-weighted NYSE/AMEX market return, or about 11.5 percent relative to S&P 500 return variance.
- **Autocorrelation evidence.** During 1985-1989, S&P 100 and S&P 500 daily returns show negative lag autocorrelations, and S&P 500 stock returns show negative cross-autocorrelations with earlier S&P 500 index returns.
- **Takeaway.** Index trading improves marketability of the stock basket, but the paper argues that it also creates microstructure price effects that bias beta estimates and add excess volatility.

2 Data and measurement

2.1 S&P 500 trading strategies

- The paper treats S&P 500 trading strategies broadly: trading that involves simultaneous or near-simultaneous trading of S&P 500 stocks or claims on the S&P 500 portfolio.
- Examples include index futures, futures options, S&P 100 and S&P 500 index options, portfolio insurance, index arbitrage, and program trades.
- S&P 500 futures were introduced in 1982; S&P 100 and S&P 500 index options began in 1983. Trading volume in these products rises sharply through the 1980s.

Interpretation: The time-series growth in index-related trading creates the historical comparison: 1975-1979 is a low-index-trading benchmark, 1980-1984 is an intermediate period, and 1985-1989 is the high-index-trading period.

2.2 Stock addition sample

- The main event sample consists of 329 stocks added to the S&P 500 during 1975-1989.
- Stocks are excluded when inclusion is tied to a reorganization of an existing S&P 500 company, when pre-inclusion CRSP data are insufficient, or when the event is contaminated by NYSE listing/delisting around inclusion.
- Daily betas are estimated over 250 trading days before and after inclusion, excluding the two days immediately around inclusion.
- Weekly betas are estimated over 50 weeks before and after inclusion.
- October 1987 is excluded to avoid the extreme market-crash observation.

Interpretation: The addition sample asks whether a stock's measured beta changes when it becomes part of the actively traded S&P 500 basket.

2.3 Matched-pair seasoned sample

- The paper also constructs 79 matched pairs of seasoned S&P 500 and non-S&P 500 stocks.
- Both stocks must be continuously listed on the NYSE from 1975 to 1989.
- One stock is in the S&P 500 before 1975, and the other is never included.
- The matched stock has the same two-digit industry code and similar market value in July 1982.

Interpretation: The matched sample tests whether the beta gap is present even without the event of inclusion. It also permits monthly beta comparisons over long windows.

2.4 Beta measurement

- Stock and market returns are measured as log returns.
- The market proxy is the CRSP value-weighted portfolio of NYSE and AMEX stocks.
- Betas are primarily OLS market-model betas estimated at daily, weekly, or monthly horizons.
- The paper also reports Scholes-Williams and Cohen et al. beta estimators to address nonsynchronous trading and other microstructure frictions.

Interpretation: OLS betas are the object that applied researchers often use; adjusted betas test whether the beta changes survive common corrections for stale prices.

2.5 Trading volume and turnover

- Trading volume is measured as average daily shares traded around index inclusion.
- Turnover is measured as dollar volume divided by the end-of-period market value.
- Stocks are sorted by pre-inclusion trading volume and turnover to test whether beta changes depend on liquidity.

Interpretation: Nonsynchronous trading predicts that beta changes should be related to before-versus-after trading frequency. Price pressure predicts an increase in beta after inclusion even in cases where trading frequency logic predicts a decrease.

2.6 Autocorrelation evidence

- Index autocorrelations are measured for S&P 100, S&P 500, and CRSP equal-weighted returns over 1975-1979, 1980-1984, and 1985-1989.
- Cross-autocorrelations measure the correlation between current stock returns and earlier S&P 500 index returns.
- Price pressure that reverses over the next few days should generate negative own autocorrelations in index returns and negative cross-autocorrelations for S&P 500 stocks.

Interpretation: Autocorrelation tests are important because beta changes alone are consistent with both price pressure and reduced nonsynchronous trading.

3 Models and empirical specifications

3.1 Baseline market model

Let r_{jt} be the log return on stock j , and let the value-weighted market return be

$$r_{mt} = \sum_{j=1}^J w_j r_{jt}. \quad (1)$$

The market model is

$$r_{jt} = \alpha_j + \beta_j r_{mt} + \epsilon_{jt}. \quad (2)$$

Interpretation: If prices measure fundamentals without microstructure effects, OLS betas estimated from this equation are interpreted as true market betas.

3.2 Basket price pressure

Suppose the first J_1 stocks are the S&P 500 basket. The observed return on each basket stock includes a common price-pressure term η_t :

$$\hat{r}_{jt} = r_{jt} + \eta_t, \quad j = 1, \dots, J_1. \quad (3)$$

Let $W = \sum_{j=1}^{J_1} w_j$ be the cumulative market weight of the basket. Then the measured market return is

$$\hat{r}_{mt} = r_{mt} + W\eta_t. \quad (4)$$

Interpretation: A market-wide basket shock enters the market proxy. This is what makes betas of basket and nonbasket stocks move in opposite directions.

3.3 Nonbasket beta bias

Define

$$R = \frac{\text{Var}(W\eta_t)}{\text{Var}(\hat{r}_{mt})}. \quad (1)$$

If η_t is uncorrelated with true returns, the measured beta of a nonbasket stock is

$$\hat{\beta}_{j,nb}^{OLS} = (1 - R)\beta_j, \quad j = J_1 + 1, \dots, J. \quad (5)$$

Interpretation: Non-S&P 500 betas are understated because the denominator of the market beta includes price-pressure variance that is not covarying with the nonbasket stock.

3.4 Basket beta bias

For a basket stock,

$$\hat{\beta}_{j,b}^{OLS} = (1 - R)\beta_j + \frac{R}{W}, \quad j = 1, \dots, J_1. \quad (7)$$

Thus, adding a stock to the basket changes its measured beta by approximately

$$\Delta \hat{\beta}_j = \frac{R}{W}. \quad (2)$$

Interpretation: Inclusion in the S&P 500 makes the stock load on the common basket price-pressure component, increasing its covariance with the measured market.

3.5 Cross-sectional compression of betas

Equations (5) and (7) imply that beta spreads within both the basket and nonbasket groups shrink by a factor of roughly $1 - R$.

Interpretation: Price pressure can make S&P 500 betas look more similar to one another and non-S&P 500 betas look more similar to one another, even if true betas have not changed.

3.6 Autocorrelation prediction

If price pressure reverses over a few days, then η_t is negatively autocorrelated. For basket stocks j and k ,

$$\text{Cov}(\hat{r}_{jt}, \hat{r}_{k,t-\ell}) = \text{Cov}(r_{jt}, r_{k,t-\ell}) + \text{Cov}(\eta_t, \eta_{t-\ell}). \quad (8)$$

Interpretation: If the true cross-autocorrelation is small, the reversal of price pressure makes observed cross-autocorrelations negative.

3.7 Nonsynchronous-trading hypothesis

The competing Scholes-Williams logic is:

- If a stock trades less frequently than the market, increasing its trading frequency raises its measured beta.
- If a stock trades more frequently than the market, increasing its trading frequency can lower its measured beta.
- Therefore, after S&P 500 inclusion, beta changes should depend on whether trading volume rises or falls and whether the stock's trading frequency is below or above the market average.

Interpretation: The core empirical separation is that price pressure predicts beta increases after inclusion in every case, while nonsynchronous trading predicts decreases in some cases.

3.8 Excess volatility calculation

For the subset where nonsynchronous trading predicts a beta decrease, the paper interprets the observed increase of 0.113 as price pressure. With $W \approx 0.75$,

$$R \approx W \times 0.113 \approx 0.085. \quad (9)$$

Interpretation: The paper infers that about 8.5 percent of daily market variance is attributable to S&P 500-trading-induced price pressure.

3.9 Cohen et al. beta adjustment

If price pressure or stale-price effects persist for up to n periods, the Cohen et al. adjusted beta combines leads and lags of market returns:

$$\hat{\beta}_{j,coh} = \frac{\sum_{\tau=1}^n \hat{\beta}_{j,ols}^{-\tau} + \hat{\beta}_{j,ols}^0 + \sum_{\tau=1}^n \hat{\beta}_{j,ols}^{+\tau}}{1 + \sum_{\tau=1}^n \hat{\rho}_m^{-\tau} + \sum_{\tau=1}^n \hat{\rho}_m^{+\tau}}. \quad (11)$$

Interpretation: Scholes-Williams is the special case $n = 1$; OLS is $n = 0$. These estimators can correct microstructure biases, but they can be very noisy in short samples.

4 Identification and empirical design

4.1 What identifies the beta effect

- The addition sample uses the same stock before and after S&P 500 inclusion.
- Inclusion dates are staggered over time, reducing the chance that all beta changes are driven by one common calendar event.
- The pre- and post-periods exclude the immediate two-day inclusion window to avoid one-time index fund purchases.
- The matched-pair sample compares S&P 500 stocks to similar non-S&P 500 stocks in the same industry and similar size.

Interpretation: The design is not randomized, but it uses within-stock changes, cross-period comparisons, and matched pairs to separate index membership from general stock characteristics.

4.2 Why 1985-1989 is central

- Index-related derivative trading becomes large in the mid-1980s.
- Portfolio insurance and program trading become more important.
- The predicted microstructure effects should therefore be strongest in 1985-1989.
- The paper compares 1985-1989 to the pre-derivative-trading and early-derivative-trading periods.

Interpretation: If the mechanism is S&P 500 basket trading, the beta changes should grow with the popularity of the basket-trading technologies.

4.3 How the paper distinguishes two mechanisms

- Both price pressure and reduced nonsynchronous trading predict higher betas for many S&P 500 additions.
- The paper classifies stocks by whether trading volume rises or falls and whether volume is above or below the market average.
- Cases 1 and 4: both hypotheses predict a beta increase.
- Cases 2 and 3: price pressure predicts an increase, but nonsynchronous trading predicts a decrease.
- The key finding is that betas still rise in cases 2 and 3.

Interpretation: This classification is the paper's main identification strategy for distinguishing price pressure from a pure stale-price story.

4.4 Additional credibility checks

- Betas are measured in monthly and quarterly event windows around inclusion to check timing.
- Daily, weekly, and monthly beta comparisons show that the bias is not only a one-day phenomenon.
- Index autocorrelations and stock-index cross-autocorrelations test for price-pressure reversals.
- Cohen et al. beta estimates test whether standard microstructure corrections eliminate the effect.

Interpretation: The results are strongest when multiple signatures line up: beta increases, beta-spread compression, negative autocorrelation, and persistence after microstructure adjustment.

4.5 Limitations

- S&P 500 additions are not random events.
- Trading volume is an imperfect proxy for nonsynchronous trading and liquidity.
- The paper uses daily and lower-frequency data, not transaction-level order-flow data.
- The price-pressure estimate of 8.5 percent relies on the model and on interpreting the residual beta increase as price pressure.
- True beta could change for some firms around inclusion, although the paper argues that immediate changes in real economic exposure are unlikely.

Interpretation: The empirical case is cumulative rather than a single clean experiment.

5 Tables and figures

5.1 Table 1 and Table 2: growth of S&P 500 trading and index autocorrelations

Read:

- Table 1 shows that S&P 500 derivative trading volume grows rapidly after futures and options become available. By the mid-1980s, derivative volumes are large relative to cash equity trading.
- Table 2 shows that S&P 100 and S&P 500 autocorrelations are positive earlier but become mostly negative in 1985-1989. The first five autocorrelations sum to -0.162 for S&P 100 and -0.107 for S&P 500.

Economic message: The institutional background and autocorrelation evidence point in the same direction: index trading becomes big enough to create temporary price effects.

Table 1
Trading volume in S&P 500 stocks and derivative products from 1982 to 1989

Year	S&P 500 trading volume ^{1,2}						S&P 100 trading ^{1,2} volume		NYSE trading ^{3,4} volume	
	Index futures		Index options		Futures options		Index options		Stocks	
	Con- tracts	Dollars	Con- tracts	Dol- lars	Con- tracts	Dol- lars	Con- tracts	Dollars	Round lots	Dollars
1982	2,935	161	—	—	—	—	161	—	16,670	495
1983	8,069	678	14	1	281	24	703	10,595	177	21,845
1984	12,364	947	12	1	673	52	1000	64,288	977	23,309
1985	15,056	1444	8	1	1090	105	1550	90,805	1686	27,774
1986	19,505	2446	1683	42	1886	237	2723	113,151	2684	36,010
1987	19,045	2895	6205	187	1877	285	3367	101,827	3044	48,143
1988	11,354	1553	4817	132	735	101	1786	57,433	1503	41,118
1989	10,560	1679	6274	199	1162	185	2063	58,371	1721	42,022

The trading volume in S&P 500 stocks is not available and therefore is substituted by the aggregate volume in all NYSE stocks obtained from the NYSE Fact Book (1991). The S&P 500 index futures and options on futures trading volumes are obtained from the Chicago Mercantile Exchange. The S&P 500 and S&P 100 index option trading volumes are obtained from a publication of the Chicago Board Options Exchange.

¹ Number of contracts is in thousands. A contract of S&P 500 index futures or futures options represents the obligation or the right to trade a dollar amount of 500 times the underlying cash index, but a contract of S&P 500 and S&P 100 index options represents the right to trade only 100 times the underlying cash index.

² Dollar volume is in billions and is calculated by multiplying the contract volume by 500 times the midyear cash index value in the case of index futures and futures options and by 100 times the midyear cash index value in the case of S&P 500 and S&P 100 index options.

³ Number of round lots in millions. Each round lot represents 100 shares.

⁴ Dollar volume in billions is obtained directly from the NYSE Fact Book (1991).

Table 1: S&P 500 and derivative-product trading volume

Table 2
Autocorrelations in index daily returns from 1975 to 1989

Period	Index	Obser- vations	Autocorrelations at lags 1 to 5				
			1	2	3	4	5
1975-1979	S&P 100	1004 ¹	0.066	0.009	0.004	-0.015	0.003
	S&P 500	1263	0.189	-0.019	0.015	0.009	0.018
	EWRETD	1263	0.451	0.119	0.121	0.137	0.154
1980-1984	S&P 100	1265	0.004	0.015	-0.024	-0.056	0.002
	S&P 500	1265	0.095	0.035	-0.019	-0.041	0.006
	EWRETD	1265	0.269	0.114	0.083	0.029	0.082
1985-1989	S&P 100	1236 ²	-0.045	-0.041	-0.030	-0.023	-0.023
	S&P 500	1236	0.024	-0.041	-0.031	-0.033	-0.026
	EWRETD	1236	0.237	0.038	0.019	-0.005	0.025

The data on the S&P 100 daily returns are obtained from the Chicago Board Options Exchange and are available only from 1976 onwards. The data on the S&P 500 daily returns are obtained from the CRSP 1989 daily index files. EWRETD stands for equally weighted returns with dividends on a portfolio of all NYSE and AMEX stocks, also obtained from the CRSP daily index files. Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987.

¹ In this row, each estimated autocorrelation has a standard error of $(1004)^{-0.5} = 0.032$. All other autocorrelations have standard errors of 0.028.

² During 1985-1989, the p -value of none of the 10 autocorrelations for the S&P 100 or the S&P 500, by itself, is less than 0.05. However, 9 of the 10 autocorrelations are negative, and 6 are at least one standard error below zero. The cumulative evidence against the null hypothesis of zero or positive autocorrelations should therefore be quite significant. (The p -value of five negative autocorrelations in S&P 100 index alone equals $0.5^5 = 0.03$.) The first five autocorrelations for the two indices sum to -0.162 and -0.107.

Table 2: daily index autocorrelations

5.2 Figure 1 and Tables 3-4: the beta-bias model and event-study betas

Read:

- Figure 1 shows the model's main prediction: basket-stock measured betas are pushed toward $(1-R)\beta + R/W$, while nonbasket betas are pushed toward $(1-R)\beta$.
- Table 3 shows the main daily addition result. During 1985-1989, daily betas rise from 0.858 to 1.069, an increase of 0.211.
- Table 4 shows the weekly version. During 1985-1989, weekly betas rise from 1.044 to 1.174, an increase of 0.130.

Economic message: The event-study results match the period when S&P 500 trading is most active. The daily effect is larger than the weekly effect, but the weekly effect remains economically meaningful.

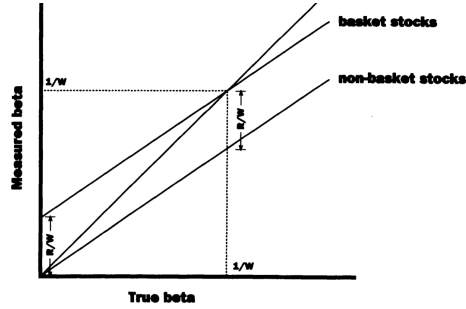


Figure 1
Price pressure and biases in S&P 500 and non-S&P 500 stock betas (also called basket and nonbasket stocks).

The measured betas of nonbasket and basket stocks in the presence of S&P 500-trading-induced price pressures are given by $\hat{\beta}_{j, \text{non}, \text{obs}} = (1 - R)\beta_j$ and $\hat{\beta}_{j, \text{basket}, \text{obs}} = (1 - R)\beta_j + R/W$, where R is the proportion of noise variance in total market portfolio variance, W is the cumulative weight of basket stocks, β_j is the true beta, and $\hat{\beta}_{j, \text{non}, \text{obs}}$ and $\hat{\beta}_{j, \text{basket}, \text{obs}}$ are the measured betas if the stock were included or excluded from the basket. The figure also illustrates that price pressure caused by S&P 500

Figure 1: beta biases under price pressure

Table 3
Daily betas before and after addition of stocks to the S&P 500 index from 1975 to 1989

Period	Number of additions to index	Average beta before inclusion	Average beta after inclusion	Increase in beta (<i>t</i> -statistic)	Number of decreases and increases in beta (<i>z</i> -statistic)
1975–1989	329	1.014	1.094	0.080 (4.30)***	142 187 (2.48)**
1975–1979	107	1.139	1.067	−0.072 (−2.75)***	68 39 (−2.80)***
1980–1984	98	1.075	1.157	0.082 (2.32)**	40 58 (1.82)*
1985–1989	124	0.858	1.069	0.211 (7.08)***	34 90 (5.03)***
1982	27	0.931	0.962	0.030 (0.56)	11 16 (0.96)
1983	10	1.027	1.072	0.044 (0.38)	5 5 (0.00)
1984	29	1.209	1.380	0.172 (2.22)**	9 20 (2.04)**
1985	27	1.013	1.004	−0.009 (−0.15)	13 14 (0.19)
1986	27	0.858	1.213	0.355 (5.96)***	4 23 (3.66)***
1987	24	0.907	1.075	0.168 (2.80)***	8 16 (1.63)
1988	22	0.773	1.062	0.288 (4.47)***	3 19 (3.84)***
1989	24	0.712	0.981	0.269 (3.78)***	6 18 (2.45)**

The sample of 329 stocks added to the S&P 500 index consists of all inclusions that satisfy the following criteria. (1) The company was newly added to the index and not the result of reorganization of an existing S&P 500 company. (2) Daily returns data are available from CRSP 1989 files for at least the 62 days preceding the following inclusion date. (3) Inclusion is not accompanied by delisting from NASDAQ and listing on the NYSE during the 62 days preceding or following the inclusion date. Betas before (after) the listing date are calculated over 250 trading days ending (starting) three days before (after) the inclusion date, using the market model with log daily returns and the OLS regression technique. The market returns are proxied by returns on a value-weighted portfolio of all NYSE and AMEX stocks. If 250 days of data are not available, the estimation interval is suitably shortened, but not to less than 60 trading days. Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987. The *t*-statistics are calculated cross sectionally using the distribution of increases in beta.

The following notation is used to denote significance levels in every table:

* Significantly different from zero at the 10 percent level.

** Significantly different from zero at the 5 percent level.

*** Significantly different from zero at the 1 percent level.

Table 3: daily betas around S&P 500 addition

Table 4
Weekly betas before and after addition of stocks to the S&P 500 index from 1975 to 1989

Period	Number of additions to index	Average beta before inclusion	Average beta after inclusion	Increase in beta (<i>t</i> -statistic)	Number of decreases and increases in beta (<i>z</i> -statistic)
1975–1989	329	1.149	1.186	0.037 (1.34)	152 177 (1.38)
1975–1979	107	1.205	1.160	−0.045 (−1.11)	57 50 (−0.68)
1980–1984	98	1.222	1.228	0.007 (0.12)	45 53 (0.81)
1985–1989	124	1.044	1.174	0.130 (2.84)***	50 74 (2.16)**

The sample of 329 stocks added to the S&P 500 index satisfies the criteria listed in Table 3. Weekly returns are calculated by aggregating the daily returns suitably. Betas before (after) the listing date are calculated over the 50 weeks ending (starting) the week before (after) the inclusion week, using the market model with log weekly returns and the OLS regression technique. If 50 weeks of data are not available, the estimation interval is suitably shortened, but not to less than 12 weeks. The market returns are proxied by returns in a value-weighted portfolio of all NYSE and AMEX stocks. Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987. The *t*-statistics are calculated cross sectionally using the distribution of increases in beta.

5.3 Table 5: short-window timing around inclusion

Read:

- Monthly betas rise around inclusion in 1985-1989.
- The average beta over the 12 months before inclusion is about 0.901, while the 12-month post-inclusion average is about 1.107.
- Quarterly windows show a similar rise from pre-event quarters to post-event quarters.

Economic message: The timing supports a microstructure interpretation rather than a slow change in fundamental risk.

Table 5
Daily betas in monthly and quarterly windows before and after addition of stocks to the S&P 500 index from 1985 to 1989

Month relative to event month	Average beta (std error)	Quarter relative to event quarter	Average beta (std error)
Event -12	0.955 (0.075)	Event -4	0.930 (0.049)
-11	0.849 (0.066)		
-10	0.946 (0.081)		
-9	0.916 (0.073)		
-8	0.859 (0.078)	-3	0.911 (0.053)
-7	0.931 (0.075)		
-6	0.872 (0.069)		
-5	0.950 (0.068)	-2	0.903 (0.048)
-4	0.901 (0.063)		
-3	0.759 (0.073)		
-2	0.875 (0.069)	-1	0.880 (0.047)
-1	0.867 (0.079)		
+1	1.005 (0.060)	+1	1.043 (0.041)
+2	1.030 (0.066)		
+3	1.035 (0.084)		
+4	1.112 (0.058)	+2	1.144 (0.043)
+5	1.145 (0.065)		
+6	1.081 (0.063)		
+7	1.276 (0.071)	+3	1.169 (0.038)
+8	1.207 (0.059)		
+9	1.060 (0.057)		
+10	1.084 (0.604)		
+11	1.092 (0.078)	Event +4	1.107 (0.040)
Event +12	1.107 (0.060)		

5.4 Tables 6-8: trading volume, turnover, and beta changes

Read:

- Table 6 shows that trading volume and turnover increase after S&P 500 inclusion, especially during the 1980s.
- Table 7 sorts by trading volume. In 1985-1989, the lowest-volume quintile has the largest beta increase, 0.366; the highest-volume quintile rises by 0.121.
- Table 8 sorts by turnover. The lowest-turnover quintile has the largest beta increase, 0.374; the highest-turnover quintile rises by 0.091.

Economic message: Low-liquidity stocks experience larger beta changes, consistent with price pressure placing more strain on stocks with thinner trading capacity.

Table 6
Daily trading volumes and annualized turnover ratios before and after addition of stocks to the S&P 500 index from 1975 to 1989

Period	Sample size	Daily trading volume ¹				Annualized turnover ratio			
		Before inclusion mean (median)	After inclusion mean (median)	Percentage increase		Before inclusion mean (median)	After inclusion mean (median)	Percentage increase	
				1	2 ²			1	2 ²
1975–1989	311	797 (465)	1068 (647)	34	20	0.61 (0.44)	0.84 (0.50)	37	33
1975–1979	95	244 (166)	275 (208)	13	−4	0.36 (0.24)	0.34 (0.26)	−5	−13
1980–1984	92	827 (469)	1163 (590)	41	16	0.70 (0.49)	1.05 (0.55)	51	46
1985–1989	124	1198 (837)	1604 (1150)	34	25	0.75 (0.62)	1.06 (0.66)	43	40

The sample of 311 stocks added to the S&P 500 index consists of all inclusions that satisfy the following criteria. (1) The company was newly added to the index and not the result of reorganization of an existing S&P 500 company. (2) Daily returns data are available from CRSP 1989 files for at least the 62 days preceding and following the inclusion date. (3) Inclusion is not accompanied by delisting from NASDAQ and listing on the NYSE during the 62 days preceding or following the inclusion date. (4) Trading volume data before and after the inclusion is available for at least 62 trading days. (Volume data is available for NYSE/AMEX stocks from July 1962 onwards, but for NASDAQ stocks from November 1982 onwards.) Before (after) trading volume is obtained by averaging the number of shares traded each day during the same estimation interval as daily betas reported in Table 3. The annualized turnover ratio is obtained by annualizing the total dollar volume during this estimation period and dividing by the end-of-period market value.

¹ In hundreds of shares.

² Two measure of percentage increase in trading volume and turnover ratio are reported. The first measure equals the ratio of average trading volume (turnover ratio) after inclusion to the average trading volume (turnover ratio) before inclusion. The second measure adjusts the after-trading volume (turnover ratio) for changes in the average trading volume (turnover ratio) of all NYSE/AMEX stocks (see Section 3.2 and note 9 for details).

Table 7
Daily betas as a function of daily trading volume from 1985 to 1989

Quintile of trading volume	Sample size	Daily trading volume ¹				Number of decreases and increases in beta		
		Before inclusion mean (median)	After inclusion mean (median)	Percentage increase		Increase in beta (t-statistic)	increases in beta (z-statistic)	
				1	2 ²			
All stocks	124	1198 (837)	1604 (1150)	34	25	0.211 (7.08)***	34	90 (5.03)***
Lowest	24	398 (372)	710 (673)	78	53	0.366 (5.64)***	4	20 (3.27)**
2	25	622 (524)	1237 (779)	99	77	0.180 (2.92)**	7	18 (2.20)**
3	25	901 (822)	1488 (1078)	65	47	0.165 (2.19)**	7	18 (2.20)**
4	25	1201 (1206)	1514 (1542)	26	22	0.230 (3.53)***	7	18 (2.20)**
Highest	25	2835 (2484)	3036 (2627)	7	5	0.121 (2.03)*	9	16 (1.40)

The sample of 124 stocks added to the S&P 500 index satisfies the criteria listed in Table 6. Betas before (after) the listing date are calculated over 250 trading days ending (starting) three days before (after) the inclusion date, using the market model with log daily returns and OLS regression technique. The market returns are proxied by returns on a value-weighted portfolio of all NYSE and AMEX stocks. When data for all 250 days are not available, the computation period is shortened to the number of days of available data, but not fewer than 60 trading days. Before (after) trading volume is obtained by averaging the number of shares traded each day during the same estimation interval as betas. The quintile ranking uses trading volume during the year before the immediate last year in order to avoid a selection bias discussed in note 10. The October 1987 data are excluded from all computations.

^{1,2} See notes to Table 6.

Table 7: betas by trading-volume quintile

Table 8
Daily betas as a function of annualized turnover ratio from 1985 to 1989

Quintile of turnover ratio	Sample size	Annualized turnover ratio				Increase in beta (t-statistic)	Number of decreases and increases in beta (z-statistic)	
		Before inclusion mean (median)	After inclusion mean (median)	Percentage increase				
				1	2 ¹			
All stocks	124	0.75 (0.62)	1.06 (0.66)	43	40	0.211 (7.08)***	34	90 (5.03)***
Lowest	24	0.30 (0.26)	0.50 (0.47)	65	62	0.374 (6.17)***	3	21 (3.67)***
2	25	0.49 (0.44)	1.22 (0.53)	148	137	0.284 (4.81)***	4	21 (3.40)***
3	25	0.68 (0.63)	0.98 (0.69)	45	46	0.209 (3.10)***	7	18 (2.20)**
4	25	0.85 (0.82)	1.06 (0.84)	25	22	0.104 (1.59)	9	16 (1.40)
Highest	25	1.39 (1.14)	1.53 (1.18)	10	10	0.091 (1.35)	11	14 (0.60)

The sample of 124 stocks added to the S&P 500 index satisfies the criteria listed in Table 6. Betas before (after) the listing date are calculated over 250 trading days ending (starting) three days before (after) the inclusion date, using the market model with log daily returns and OLS regression technique. The market returns are proxied by returns on a value-weighted portfolio of all NYSE and AMEX stocks. When data for all 250 days are not available, the computation period is shortened to the number of days of available data, but not fewer than 60 trading days. Before (after) trading volume is obtained by annualizing the total dollar volume during the same estimation interval as betas and dividing by the end-of-period market value. The quintile ranking uses turnover ratio during the year before the immediate last year in order to avoid a selection bias discussed in note 10. The October 1987 data are excluded from all computations.

Table 8: betas by turnover quintile

5.5 Table 9: separating price pressure from nonsynchronous trading

Read:

- The diagonal cases, where both hypotheses predict beta increases, show a combined beta increase of 0.241.
- The off-diagonal cases, where price pressure predicts an increase but nonsynchronous trading predicts a decrease, still show a beta increase of 0.113.
- This 0.113 increase is statistically significant and is used to infer the 8.5 percent excess-volatility estimate.

Economic message: This is the paper's strongest mechanism table. A pure stale-price story cannot explain why betas rise in the cases where it predicts a fall.

Table 9
Testing the price pressures versus the nonsynchronous-trading hypotheses from 1985 to 1989

Trading volumes	Before and after volumes were smaller than market average	Before and after volumes were greater than market average	All cases
After volume was greater than before volume	Case 1: 86 stocks ¹ 730 1257 0.250 (6.56) ***	Case 3: 3 stocks 4287 5787 -0.168 (-1.03)	89 stocks 850 1409 0.236 (6.21) ***
After volume was smaller than before volume	Case 2: 24 stocks 1399 1196 0.148 (2.76) **	Case 4: 3 stocks 5024 4038 0.006 (0.03)	27 stocks 1801 1511 0.132 (2.55) **
All cases	110 stocks 876 1243 0.227 (7.08) ***	6 stocks 4655 4912 -0.080 (-0.65)	116 stocks 1071 1433 0.211 (6.68) ***
Sample description	Predictions of the price pressures hypothesis	Predictions of the nonsynchronous-trading hypothesis	Results
Diagonal terms cases 1 and 4	Increase	Increase	89 stocks 875 1350 0.241 (6.44) ***
Off-diagonal terms cases 2 and 3	Increase	Decrease	27 stocks 1719 1706 0.113 (2.11) **

The sample of 116 stocks added to the S&P 500 index satisfies the criteria listed in Table 6, plus the additional criterion that either the before and after trading volumes are both greater than weighted-average market volume, or both are smaller. Betas before (after) the listing date are calculated over 250 trading days ending (starting) three days before (after) the inclusion date, using the market model with log daily returns and OLS regression technique. The market returns are proxied by returns on a value-weighted portfolio of all NYSE and AMEX stocks. When data for all 250 days are not available, the computation period is shortened to the number of days of available data, but not fewer than 60 trading days. Before (after) trading volume is obtained by averaging the number of shares traded each day during the same estimation interval as betas. The weighted-average market volume is calculated as described in Section 3.2. The October 1987 data are excluded.

5.6 Table 10 and Figure 2: matched-pair beta distributions

Read:

- Table 10 compares 79 pairs of seasoned S&P 500 and non-S&P 500 stocks.
- In 1985-1989, S&P 500 stocks have higher monthly, weekly, and daily betas by 0.125, 0.279, and 0.380.
- Figure 2 shows S&P 500 beta distributions shifting right and non-S&P 500 distributions shifting left in the high-index-trading period.
- The narrowing of beta distributions within groups is consistent with the beta-compression prediction of the price-pressure model.

Economic message: The beta effect is not limited to newly added stocks; it also appears in seasoned S&P 500 stocks relative to similar nonmembers.

Table 10
Matched-pairs tests of seasonal S&P 500 and non-S&P 500 stock betas from 1975 to 1989

Estimation interval	Average value of S&P 500 stock betas (cross-sectional std deviation)	Average value of non-S&P 500 stock betas (cross-sectional std deviation)	Difference in S&P 500 and non-S&P 500 stock betas (<i>t</i> -statistic)	Nonparametric tests: number of times S&P 500 beta is smaller or greater than non-S&P 500 beta (<i>z</i> -statistic)
Monthly betas				
1975-1979	1.146 (0.404)	1.118 (0.408)	0.028 (0.56)	39 40 (0.11)
1980-1984	0.965 (0.416)	0.927 (0.458)	0.038 (0.70)	35 44 (1.01)
1985-1989	1.063 (0.348)	0.938 (0.319)	0.125 (2.83)***	30 49 (2.14)**
Weekly betas				
1975-1979	1.023 (0.353)	0.956 (0.419)	0.067 (1.32)	32 47 (1.69)
1980-1984	0.873 (0.320)	0.830 (0.347)	0.043 (1.03)	34 45 (1.24)
1985-1989	1.026 (0.304)	0.747 (0.250)	0.279 (6.53)***	16 63 (5.29)***
Daily betas				
1975-1979	0.906 (0.391)	0.829 (0.463)	0.077 (1.39)	28 51 (2.59)***
1980-1984	0.802 (0.323)	0.690 (0.332)	0.112 (2.50)**	30 49 (2.14)**
1985-1989	1.011 (0.240)	0.631 (0.190)	0.380 (10.67)***	7 72 (7.31)***

The sample of 79 pairs of S&P 500 and non-S&P 500 stocks satisfies the following criteria. (1) Both stocks were listed on the NYSE continuously from 1975 to 1989. (2) The first stock was included in the S&P 500 before 1975, and the second was never included. (3) Both stocks have the same two-digit industry code. (4) The market value of the non-S&P 500 stock is no more than twice and no less than half of the market value of the S&P 500 stock on July 1, 1982 (the midpoint in time). Betas are calculated using the market model with log returns and the OLS technique. The market returns are proxied by the returns on a value-weighted portfolio of NYSE and AMEX stocks. Data

Table 10: matched-pairs beta tests

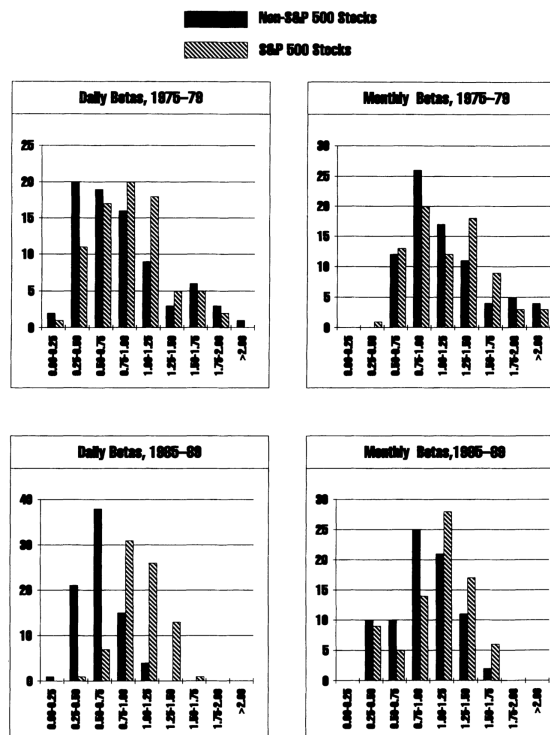


Figure 2: beta distributions

5.7 Tables 11-12: cross-autocorrelation evidence

Read:

- Table 11 shows that after inclusion in 1985-1989, cross-autocorrelations between current stock returns and earlier S&P 500 index returns become negative from lags 2 to 5.
- Table 12 shows that seasoned S&P 500 stocks have more negative cross-autocorrelations with earlier S&P 500 returns than seasoned non-S&P 500 stocks during 1985-1989.
- These patterns are much weaker or absent in earlier periods.

Economic message: Negative cross-autocorrelation is exactly what temporary price pressure predicts: the basket price effect is partly reversed over the following days.

Table 11
Cross autocorrelations between current stock returns and earlier-period S&P 500 index returns using daily data before and after addition of stocks to index from 1975 to 1989

Period	Sample size	Lag	Before addition to index		After addition to index	
			Average cross autocorrelation, cross-sectional standard deviation, number positive, and number negative		Average cross autocorrelation, cross-sectional standard deviation, number positive, and number negative	
1975–1979	107	1	0.1561 (0.1030)	98 9	0.1254 (0.0858)	100 7
		2	0.0081 (0.0632)	59 48	0.0220 (0.0475)	76 31
		3	0.0415 (0.0487)	84 23	0.0281 (0.0486)	78 29
		4	0.0255 (0.0443)	76 31	0.0061 (0.0485)	58 49
		5	-0.0097 (0.0494)	50 57	0.0091 (0.0494)	60 47
1980–1984	97	1	0.1288 (0.0958)	87 11	0.0971 (0.0892)	86 12
		2	0.0230 (0.0439)	71 27	0.0232 (0.0488)	70 28
		3	0.0130 (0.0500)	55 43	0.0209 (0.0493)	64 34
		4	-0.0047 (0.0540)	46 52	0.0121 (0.0508)	61 37
		5	0.0072 (0.0501)	53 45	-0.0037 (0.0469)	48 50
1985–1989	124	1	0.1296 (0.0719)	117 7	0.0375 (0.0779)	89 35
		2	0.0073 (0.0506)	68 56	-0.0127 (0.0549)	54 70
		3	0.0137 (0.0452)	76 48	-0.0104 (0.0475)	49 75
		4	-0.0035 (0.0426)	60 64	-0.0271 (0.0616)	38 86
		5	0.0006 (0.0467)	58 66	-0.0137 (0.0435)	45 79

The sample of 329 stocks added to the S&P 500 index satisfies the criteria listed in Table 3. Cross autocorrelations between stock returns and earlier-period S&P 500 index returns before (after) the listing date are calculated over 500 trading days ending (starting) three days before (after) the inclusion date. (A cross autocorrelation at lag 2, for example, specifies the correlation between stock returns on day t and the S&P 500 index returns on day $t - 2$.) If 500 days of data are not available, the estimation interval is suitably shortened, but not to less than 60 trading days. Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987.

Table 11: cross-autocorrelations around additions

Table 12
Cross autocorrelations between seasoned S&P 500 and non-S&P 500 stock returns and earlier-period S&P 500 index returns using daily data from 1975 to 1989

Period	Lag	79 seasoned non-S&P 500 stocks		79 seasoned S&P 500 stocks	
		Average cross autocorrelation, cross-sectional standard deviation, number positive, and number negative		Average cross autocorrelation, cross-sectional standard deviation, number positive, and number negative	
1975–79	1	0.1313 (0.0617)	74 5	0.1426 (0.0655)	78 1
	2	0.0165 (0.0358)	52 27	0.0028 (0.0356)	42 37
	3	0.0274 (0.0342)	60 19	0.0147 (0.0278)	55 24
	4	0.0201 (0.0307)	60 19	0.0252 (0.0323)	65 14
	5	0.0273 (0.0294)	65 14	0.0191 (0.0253)	59 20
1980–84	1	0.1310 (0.0606)	77 2	0.1044 (0.0635)	74 5
	2	0.0312 (0.0304)	70 9	0.0231 (0.0313)	63 16
	3	0.0173 (0.0351)	58 21	0.0065 (0.0312)	43 36
	4	-0.0008 (0.0331)	37 42	-0.0053 (0.0296)	36 43
	5	0.0101 (0.0251)	47 32	0.0068 (0.0286)	47 32
1985–89	1	0.0916 (0.0680)	72 7	0.0333 (0.0515)	57 22
	2	0.0000 (0.0342)	43 36	-0.0104 (0.0310)	28 51
	3	0.0004 (0.0304)	38 41	-0.0113 (0.0279)	27 52
	4	-0.0126 (0.0369)	31 48	-0.0136 (0.0302)	23 56
	5	0.0032 (0.0282)	40 39	-0.0115 (0.0267)	27 52

The sample of 79 pairs of seasoned S&P 500 and non-S&P 500 stocks satisfies the criteria listed in Table 10. Cross autocorrelations between stock returns and earlier-period S&P 500 index returns are calculated over the full five-year period. (A cross autocorrelation at lag 2, for example specifies the correlation between stock returns on day t and the S&P 500 index returns on day $t - 2$.) Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987.

Table 12: seasoned matched-pair cross-autocorrelations

5.8 Tables 13-14: Scholes-Williams and Cohen et al. adjusted betas

Read:

- Table 13 shows that adjusted betas for added stocks are higher than OLS betas, but the before-after gap does not vanish.
- Table 14 shows that for matched pairs in 1985-1989, daily beta differences remain positive even after using Scholes-Williams or Cohen et al. corrections.
- The adjusted estimators are noisy, especially with short samples, but they do not overturn the main conclusion.

Economic message: Standard microstructure corrections reduce some bias but cannot fully eliminate the S&P 500/non-S&P 500 beta gap in the high-index-trading period.

Table 13
Scholes and Williams (1977) and Cohen et al. (1983) betas for all the 124 stocks added to the S&P 500 index from 1985 to 1989

Number of leads and lags (n)	Cohen et al. betas before addition to index	Cohen et al. betas after addition to index	Difference in betas (t -statistic)
Weekly betas			
0 OLS	1.044	1.174	0.130 (2.84)***
1 SW	1.262	1.360	0.098 (0.42)
Daily betas			
0 OLS	0.858	1.069	0.211 (7.08)***
1 SW	1.030	1.130	0.100 (2.71)***
2	1.102	1.136	0.033 (0.70)
3	1.139	1.164	0.025 (0.49)
4	1.151	1.262	0.110 (1.55)
5	1.195	1.299	0.104 (1.25)

The sample of 329 stocks added to the S&P 500 index satisfies the criteria listed in Table 3. Betas before (after) the listing date are calculated over 250 trading days ending (starting) three days before (after) the inclusion date. If 250 days of data are not available, the estimation period is suitably shortened, but not to less than 60 trading days. The market returns are proxied by returns on a value-weighted portfolio of all NYSE and AMEX stocks. Cohen et al. betas are obtained as described in Section 6. All betas are calculated with log stock and log market returns. Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987.

Table 13: adjusted betas for added stocks

Table 14
Scholes and Williams (1977) and Cohen et al. (1983) betas for the 79 matched pairs of seasoned S&P 500 and non-S&P 500 stocks from 1985 to 1989

Number of leads and lags (n)	Cohen et al. beta of non-S&P 500 stocks	Cohen et al. beta of S&P 500 stocks	Difference in betas (t -statistic)
Monthly betas			
0 OLS	0.938	1.063	0.125 (2.83)***
1 SW	1.005	0.970	-0.035 (0.42)
Weekly betas			
0 OLS	0.747	1.026	0.279 (6.53)***
1 SW	0.933	0.018	0.085 (0.42)
Daily betas			
0 OLS	0.631	1.011	0.380 (10.66)***
1 SW	0.769	1.020	0.251 (6.42)***
2	0.789	1.035	0.246 (5.73)***
3	0.799	1.029	0.229 (4.98)***
4	0.800	1.049	0.249 (4.83)***
5	0.798	1.044	0.246 (4.53)***

The sample of 79 pairs of seasoned S&P 500 and non-S&P 500 stocks satisfies the criteria listed in Table 10. The market returns are proxied by returns on a value-weighted portfolio of all NYSE and AMEX stocks. Cohen et al. betas are obtained as described in Section 6. All betas are calculated with log stock and log market returns using the entire five years of data. Data for the month of October 1987 are excluded to remove the period affected by the market crash of 1987.

Table 14: adjusted betas for matched pairs

6 Short summary

- Vijn studies whether S&P 500 trading strategies bias measured stock betas.
- The price-pressure hypothesis says basket trading creates temporary common price shocks in S&P 500 stocks.
- The nonsynchronous-trading hypothesis says higher trading frequency makes S&P 500 stock prices less stale.
- Both hypotheses predict higher measured betas for many S&P 500 additions, but they differ in cross-sectional predictions.
- The paper studies 329 S&P 500 additions between 1975 and 1989.
- Daily betas rise sharply for additions during 1985-1989, the period of intense S&P 500 derivative and program trading.
- Weekly betas also rise, implying that the effect is not purely intraday.
- Trading volume and turnover rise after inclusion, but low-liquidity stocks show the largest beta increases.
- In the subset where nonsynchronous trading predicts beta decreases, betas still rise by 0.113.
- This residual increase implies price pressure equal to about 8.5 percent of daily variance in the CRSP value-weighted market return.
- A matched sample of seasoned S&P 500 and non-S&P 500 stocks confirms higher S&P 500 betas in 1985-1989.
- Index autocorrelations and stock-index cross-autocorrelations become negative in the high-index-trading period.
- Scholes-Williams and Cohen et al. corrections reduce some microstructure concerns but do not eliminate the main effect.
- The central takeaway is that market-wide basket trading can distort measured systematic risk.

7 Conclusion

7.1 Core conclusion

Vijn argues that S&P 500 trading strategies generate liquidity-related price pressure that biases measured betas. S&P 500 stock betas are overstated and non-S&P 500 stock betas are understated, especially during the high-index-trading period of 1985-1989.

7.2 Economic intuition

Basket trading makes many S&P 500 stocks move together for liquidity reasons. Because the market proxy itself contains these price-pressure movements, basket stocks appear to covary more with the market, while nonbasket stocks appear to covary less. When price pressure reverses, negative autocorrelations and cross-autocorrelations appear.

7.3 Takeaway

The paper shows that beta is not purely a fundamental risk measure when market-index trading is large relative to liquidity. Index trading can improve tradability of the market portfolio, but it may also create excess volatility and contaminate standard risk estimates used in asset pricing and capital budgeting.